SF SHOW

OPERATOR'S MANUAL

THE SHOW TEMPLATE, THE E16 SCENE AND THE OXI ONE MKII
ONE LIVE SET ONE E16 SCENE ONE SEQUENCER

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STRANGE FREQUENCIES

E16 PAGES, PRESSES AND ADDRESSES IN THIS MANUAL ARE GENERATED FROM CONTROLLERS/E16/LAYOUT.PY.

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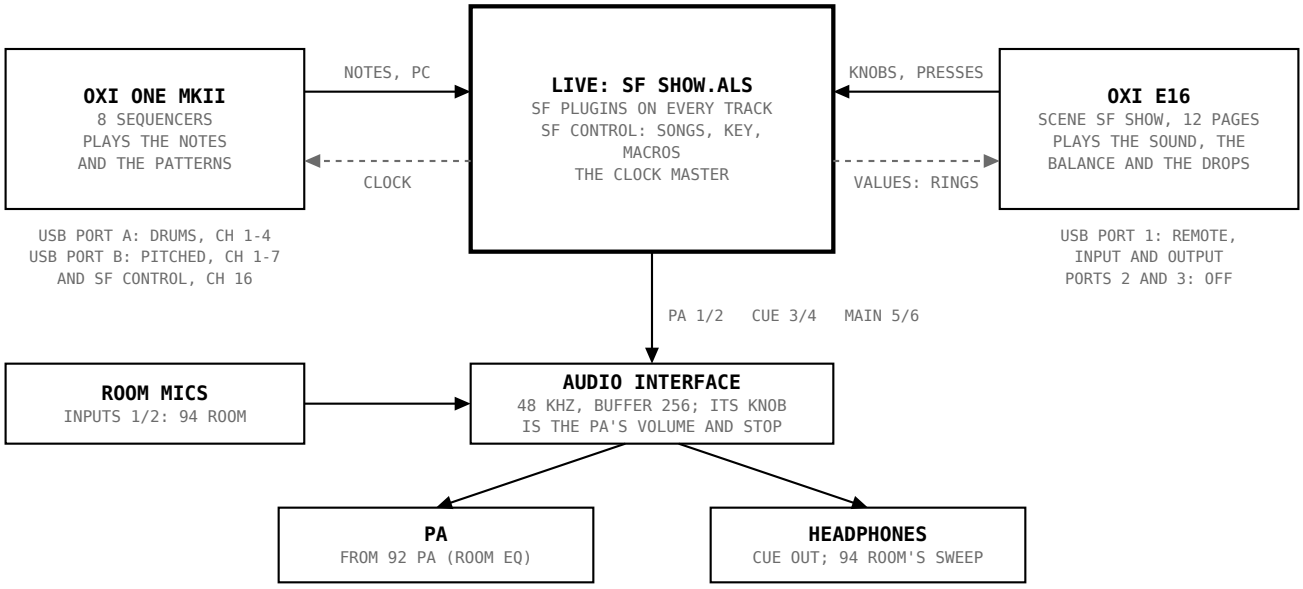
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1 THE RIG

Three devices, three jobs. The OXI One MkII plays the notes and the patterns. Live, running SF SHOW.als, makes the sound. The E16 plays the sound, the balance and the drops. Nothing on stage is loaded, routed or mapped: everything below is set once.



DEVICE	ITS JOB
OXI ONE MKII	Every note, rhythm and pattern change, and each song's Program Change.
LIVE: SF SHOW.ALS	Every sound (one SF plugin per role), the songs, the master chain, the recording, the clock.
OXI E16	Energy, build and drop, the volumes; then one instrument, the returns or the macros per page.
AUDIO INTERFACE	Its output knob is the PA's volume and the emergency stop: the room goes quiet, the recording carries on.

THE TWO HANDS

— **Left hand: rows 1 and 2.** The three group ENERGY knobs, BUILD (press it: DROP) and the four main volumes. They are the same on every page, so the arrangement is always under one hand.

— **Right hand: rows 3 and 4.** The page: one instrument, the returns, or the macros. Change pages to go deep into one sound while the left hand keeps the set moving.

WHAT NEVER CHANGES IN A SHOW

— The tracks, the routing, the E16 scene and the mappings. A song is a stored set of values in every SF plugin, recalled on the bar: nothing is loaded between tracks.

2 THE SHOW TEMPLATE

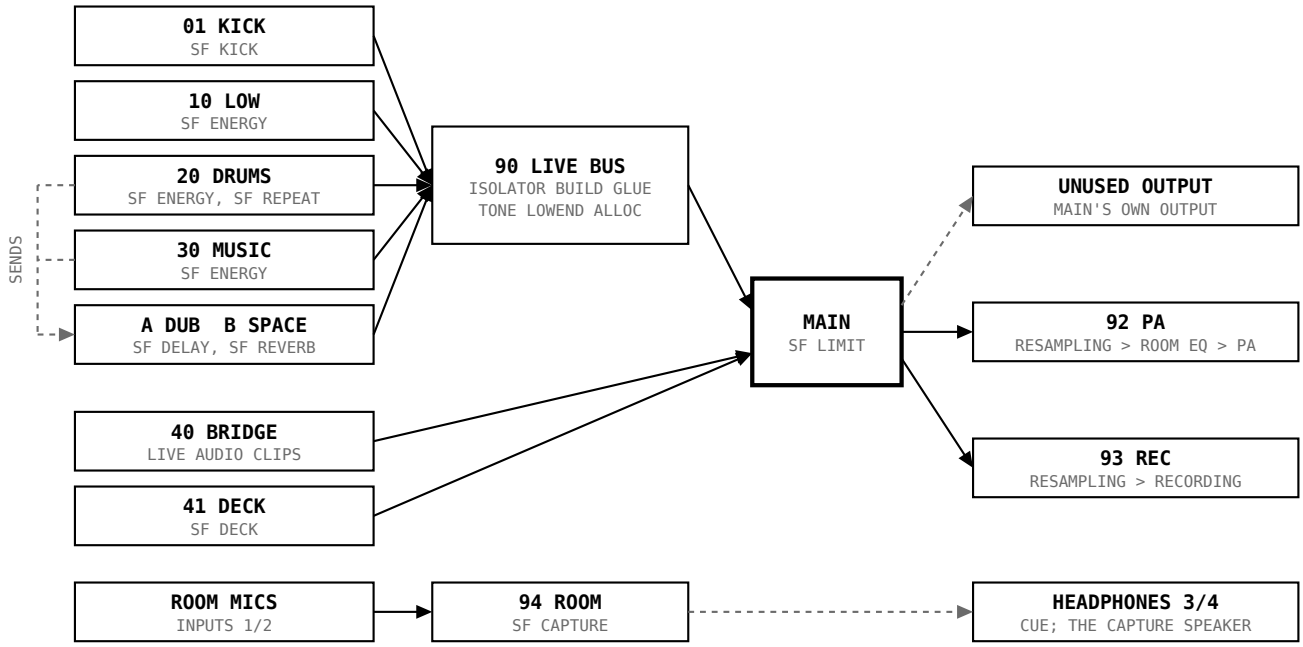
SF SHOW.als is built: every track, group and return, every SF plugin with its starting settings, the routing, every instrument's MIDI From on its OXI One port and channel, and every E16 address mapped. Set Live's own settings once (below), open it, and keep it as the template for every show.

TRACKS

TRACK	DEVICES	MIDI FROM	OUT
00 CONTROL	SF CONTROL	ONE port B, ch 16	Main
01 KICK	SF KICK	ONE port A, ch 1	90 LIVE BUS
10 LOW (group)	SF ENERGY, PROFILE BASS		90 LIVE BUS
11 BASS	SF BASS	port B, ch 1	LOW
12 RUMBLE	SF RUMBLE, SOURCE KICK BUS	none: it follows the kick	LOW
20 DRUMS (group)	SF ENERGY, PROFILE DRUMS; SF REPEAT		90 LIVE BUS
21 CLOSED	SF PERC: closed hat	port A, ch 2	DRUMS
22 OPEN	SF PERC: open hat or ride	port A, ch 3	DRUMS
23 CLAP	SF PERC: clap or rim	port A, ch 4	DRUMS
24 STRIKE	SF STRIKE	port B, ch 6	DRUMS
30 MUSIC (group)	SF ENERGY, PROFILE MUSIC		90 LIVE BUS
31 ACID	SF LEAD: ACID 303-TYPE	port B, ch 2	MUSIC
32 STAB	SF LEAD: STAB DUB	port B, ch 3	MUSIC
33 TEXTURE	SF PAD: GRANULAR	port B, ch 4	MUSIC
34 VOICE	SF VOICE	port B, ch 5	MUSIC
40 BRIDGE	Live audio clips		Main
41 DECK	SF DECK	port B, ch 7 (hot cues; optional)	Main
A DUB (return)	SF DELAY, DRY off		90 LIVE BUS
B SPACE (return)	SF REVERB, DRY off		90 LIVE BUS
90 LIVE BUS	SF ISOLATOR > BUILD > GLUE > TONE > LOWEND; SF ALLOC (off at first)	sends disabled	Main
Main	SF LIMIT		5/6: a pair that goes nowhere
92 PA	SF ROOM EQ > SF LIMIT (safety)	audio: Main, Post Mixer	the PA outputs (1/2)
93 REC	nothing	audio: Main, Post Mixer; Monitor Off	Sends Only
94 ROOM	SF CAPTURE	audio: Ext. In 1/2, the room mics; Monitor In	the headphones (3/4)

Every instrument track is on Monitor In: it plays from the ONE whether or not it is armed. MIDI From is set on every instrument: in Live the ONE's port A is OXI ONE MKII (Port 1), port B is (Port 2). The kick stays outside LOW, so the bass and the rumble fade together while the kick holds the floor.

ROUTING



— Main is the set, after SF LIMIT: exporting it gives the release mix, and its own output goes nowhere. 92 PA and 93 REC both take the Main after its fader. The venue's correction (ROOM EQ) exists only on 92 PA, so the recording never hears it.

— 92 PA's SF LIMIT is a safety: CEILING -0.5 dB, no clip, AUTO off. The set's peaks are already at -1 dBTP, so it only catches what the room EQ's phase pushes past. Keep 92 PA's fader at 0 dB: the PA's level is the interface's.

— Never send 90 LIVE BUS to 92 PA: BRIDGE, DECK and SF LIMIT would be left out of the PA, and 93 REC would record nothing of the set.

— **Every send on 90 LIVE BUS is disabled.** The returns feed LIVE BUS, and Live can't compensate a return's delay on a track whose matching send is active, even at zero.

— Pull 90 LIVE BUS down and the live instruments and their effects go while BRIDGE and DECK play on. Turn the interface's output down and the room goes quiet while REC keeps recording.

— 94 ROOM is the room mics, a stereo pair. Armed, it records them with the set. Its SF CAPTURE never passes them (the set never hears the mics): it only plays SF TERROIR's sweep, to a speaker on the headphone out, when you capture the room (chapter 8).

LIVE'S OWN SETTINGS, ONCE PER COMPUTER

PORT (SETTINGS > LINK, TEMPO & MIDI)	TRACK	SYNC	REMOTE
OXI ONE MKII (Port 1) and (Port 2), input	on	off	off
OXI ONE MKII (Port 1), output	off	on	off
OXI E16 (Port 1), input and output	off	off	on
every other port	off	off	off

— Takeover Mode: Value Scaling. Audio: 48 kHz, buffer 256. These live in Live's preferences, not in any set: nothing in SF SHOW.als can set them.

OPEN IT, CHECK, SAVE

1. **MIDI From** is set on every instrument, as in the table above (chapter 3 shows the ONE's side). If your Live names the ONE differently, a track shows its port as missing: set it by the table. Never All Ins: every instrument would play every part.
2. **Inputs and outputs**, set for the PA on 1/2 and headphones on 3/4. 92 PA to the pair that feeds the PA (1/2 in the set). Cue Out, on the Main track, to the headphones (3/4), and 94 ROOM there too. Main to a pair that goes nowhere (5/6). Check all three before any sound reaches a PA: the PA pair carries 92 PA and nothing else. The room mics on inputs 1 and 2 (1: the mic on your left as you face the speaker), and in Live's Settings > Audio > Input Config, 1/2 as a stereo pair: 94 ROOM's input.
3. **Two options:** Delay Compensation on; Reduced Latency When Monitoring off.
4. **Save** as a template (File > Save Live Set As Template) or into your show folder. MIDI From and the mappings belong to the set, so every set made from it starts ready.

THE MASTER CHAIN

— Start GLUE, TONE and LOWEND gently: the set doesn't need every processor working all the time.

— Soundcheck the busiest passage of the night and set SF LIMIT's INPUT for 1-3 dB of limiting there. LIMIT AUTO on, AUTO MODE DOWN ONLY, for shows: it holds the set at TARGET and holds through breakdowns.

— SF ALLOC starts off. Turn it on band by band once the arrangement is known, and judge it by ear.

3 THE OXI ONE MKII

The ONE plays every part. Eight sequencers, one per role; port A carries the drums, port B everything pitched and SF CONTROL. Live is the clock: the ONE follows it.

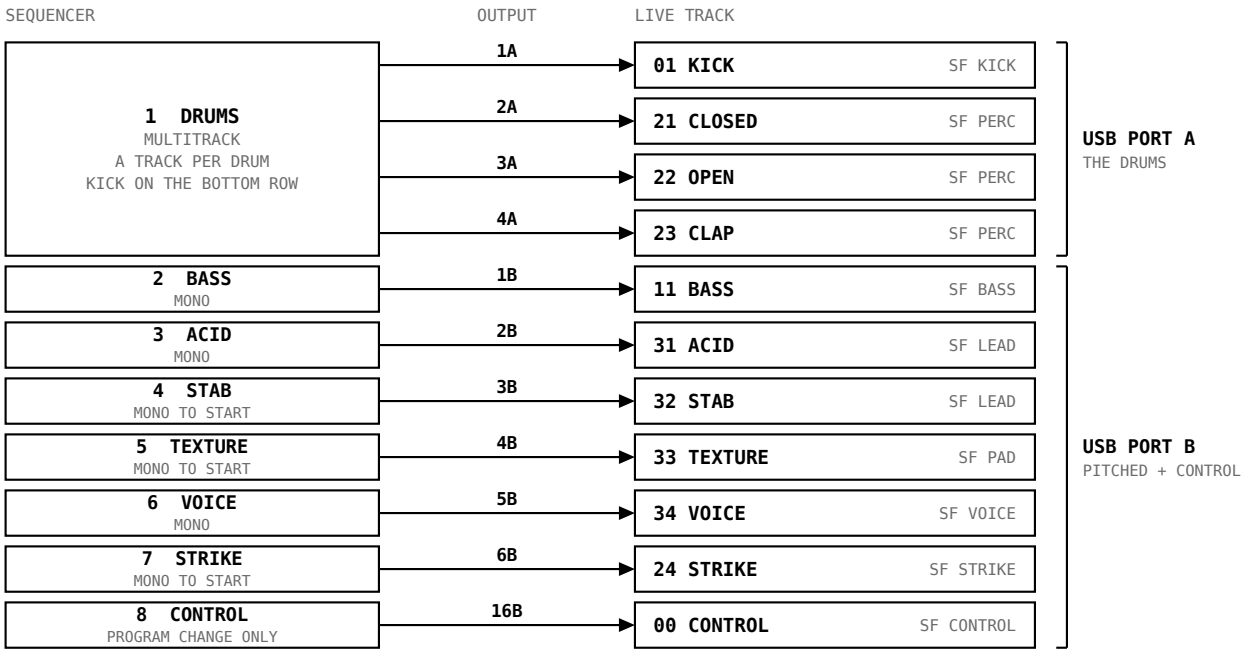
WHAT THE ONE BRINGS

	OXI ONE MKII
SEQUENCERS	8, each in one of six modes: Mono, Poly (up to 7 voices), Chord (up to 8 voices), Multitrack (8 tracks), Stochastic, Matriceal.
PATTERNS	Up to 128 steps; 128 patterns per project.
USB	USB-C for MIDI and power: 32 MIDI channels, as two ports in Live (banks A and B, 16 channels each).
UNUSED HERE	TRS MIDI in and out, Bluetooth MIDI, 8 CV and 8 gate outputs, clock in and out. The rig runs on USB alone.

Sources: Thomann, OXI One MkII; Milk Audio, OXI ONE MkII specifications; Elektronauts: banks A and B appear as two USB MIDI devices. Check menus and limits against your firmware's manual.

SEQUENCERS, OUTPUTS, TRACKS

On the ONE an output is a channel and a port: 1A is channel 1 on port A, 16B channel 16 on port B. Set each sequencer's output as below, and each Live track's MIDI From to the same port and channel.



— **KICK lane on C1.** With KEY TRACK on, any other note transposes the kick (C1 = TUNE), so C1 is right either way. PERC plays any note.

— **RUMBLE needs no lane:** SOURCE KICK BUS follows SF KICK.

— **STAB:** if the ONE sends whole chords (Chord or Poly mode), set CHORD on the plugin to NONE; if it sends single notes, CHORD voices them. Not both.

— **ACID** slides when notes overlap, over 60 ms like a 303: write legato notes for slides.

— **VOICE:** C3 plays slice 1 of the loaded phrase, C#3 slice 2, and so on; below C3 the whole phrase.

— **DECK** takes hot cues on port B, channel 7 (C1-D#1 deck A, C2-D#2 deck B): any spare lane set to 7B can cue it.

CLOCK AND PORTS

— **Live is the clock master.** On the ONE: sync USB (stopped, Shift + Stop until the header shows USB, not INT); in its config (Shift + Back), Start Stop Msgs Ignore No, MIDI Transport Msgs Send Off, MIDI Thru off, Auto Sync on (it starts with Live). Never send clock both ways.

— **Live sends its clock only while it plays.** Start and stop from Live, never with the ONE's Play: started on its own, the ONE runs at its own tempo and ignores Live's.

— **Audio:** 48 kHz, buffer 256 (Pocket needs 512 or less), launch quantization 1 bar, record quantization off.

— **Calibrate once:** record a plain quarter-note pattern from the ONE, compare it with Live's grid and set the ONE port's MIDI Clock Sync Delay in Live's preferences. Leave POCKET OFFSET on SF CONTROL at 0 unless the pump is still off after that.

LIVE MIDI PORT	TRACK	SYNC	REMOTE
OXI ONE MKII (Port 1), in: the ONE's port A	on	off	off
OXI ONE MKII (Port 2), in: its port B	on	off	off
OXI ONE MKII (Port 1), out: the clock	off	on	off
OXI E16 (Port 1), in and out	off	off	on
Every other port, E16 Ports 2 and 3 included	off	off	off

PROJECTS AND PATTERNS

— **One ONE project per track** (per piece of music), with a pattern per section: SPARSE, MAIN, VARIATION, EXIT.

— **Every project starts as a copy of SF START**, the startup project: every sequencer set as above, and four scenes of simple patterns in columns 1-4, SPARSE, MAIN, VARIATION, EXIT (controllers/oxi-one: SFSTART.oxipro and its card). Save, an empty slot, Duplic proj here; then sequencer 8's Program Change, the song's number: PC n calls SONG n. check_project.py checks the copy.

— **Sequencer 8 carries the song.** The ONE sends a Program Change (and bank) per pattern: give sequencer 8's patterns the track's song number, output 16B. Loading the project recalls the matching SF song on the bar (chapter 6). Program Change stays off on the musical sequencers.

— **When patterns change is set on the ONE** (its launch and arranger settings): Live's launch quantization doesn't reach it.

— **KEY on SF CONTROL starts OFF:** the ONE's notes set the pitch, and KICK and RUMBLE are tuned per song with TUNE. Turn KEY on for songs that should change key without reprogramming the ONE.

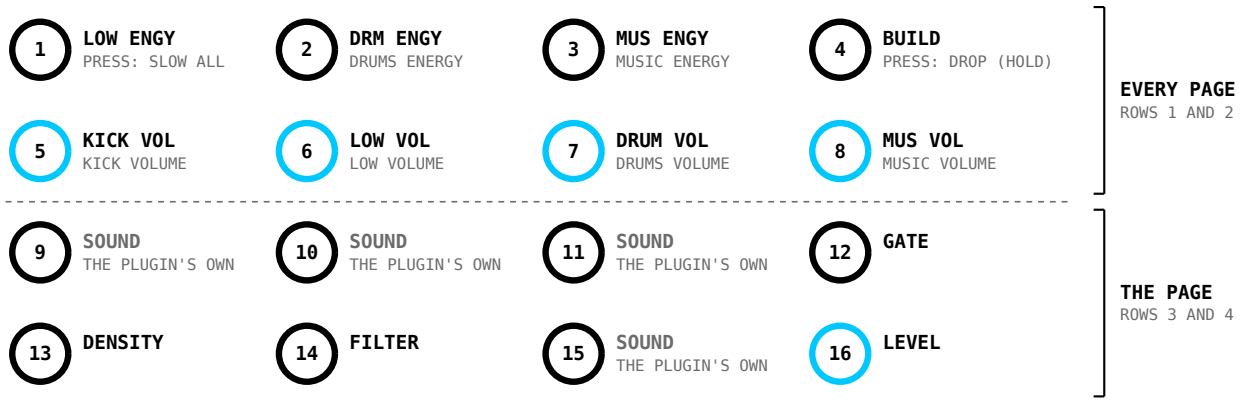
— **Taking the kick out:** mute its lane on the ONE. Pulling the KICK fader down leaves it playing on the bus, so Pocket and RUMBLE keep pumping to a silent kick (which can be the point).

4 THE E16 SCENE

One E16 scene runs the whole show: SF SHOW, twelve pages of sixteen knobs. Rows 1 and 2 are the same on every page; rows 3 and 4 are the page. The set comes with every address mapped.

THE GRID

The E16 is a 4 × 4 grid of endless encoders, numbered row by row from the top left. Its screen shows each knob's short name (below: LOW ENGY, DRM ENGY ...).



ROWS 3 AND 4 AS ON AN INSTRUMENT PAGE. LIGHT BLUE RING: A VOLUME.

— **1-3, the group ENERGY knobs:** centre is neutral, left takes energy out, right puts it in (filter, drive, space, echo and chop together).

— **Press 1: SLOW ALL.** A quick press flips knob 1 to SLOW ALL on SF CONTROL (the ring turns green): turn it to set how long your next moves take, OFF (at once) to 128 BAR. Press again: LOW ENERGY. Knobs already travelling keep going.

— **4, BUILD:** turn it up over a build. **Press it: DROP**, on every page. The drop lands on the next bar line (chapter 7).

— **5-8, the volumes:** KICK, LOW, DRUMS, MUSIC. Each is mapped with a maximum of 0 dB, so a full turn never runs past the soundcheck balance.

— **On an instrument page** 13 is DENSITY, 14 FILTER and 16 LEVEL, and 12 is GATE wherever there is one: the hand learns one layout. The rest is the plugin's own performance row.

RINGS, PAGES, PRESSES

— **Rings:** light blue is a volume, white is everything else, green is knob 1 on SLOW ALL or a knob travelling (knob 1 itself never shows a glide: it swaps, so it always shows the value of the side it is on). Live sends every value back, so the rings follow song recalls and mouse moves.

— **A recalled knob is yours the moment you turn it.** With Takeover Mode Value Scaling, a knob that disagrees with Live (after a song recall) moves the sound smoothly from where it is, never jumps.

— **Changing pages sends nothing.** A knob on two pages (LOW ENERGY is on all twelve) sends the same address from both: two ways to one mapping.

— **A hold press** is a CC, 127 on press and 0 on release: the parameter is on for as long as the knob is down (DROP, THROW, ROLL). **A toggle press** is a note: Live flips the switch on every Note On (FREEZE), so the E16 can never think FREEZE is on while Live has it off.

— Every press is one the set plays through if it happens by accident (a throw, a roll, a freeze, a drop). Nothing that would stop the set is on the E16: that is the interface's output knob.

— **On screen, the same grid.** Every SF plugin's window starts with its page of this scene: its own encoders lit where they sit on the E16, in these colours, the rest dark, its presses under their knobs. Several windows side by side read like the E16 itself. LEAD, PERC and ENERGY are on several tracks: their windows find their encoders by the track's name (VST3).

WHERE	KNOB	PRESS	MESSAGE	LIVE TARGET
every page	1	SLOW ALL, swap	turn 2: CC 34 ch 16	00 CONTROL > SF CONTROL > SLOW ALL
every page	4	DROP, hold	CC 21 ch 15	90 LIVE BUS > SF BUILD > DROP
SPACE	9	THROW, hold	CC 24 ch 12	A DUB > SF DELAY > THROW
SPACE	10	FREEZE, toggle	note 25 (C#0) ch 12	A DUB > SF DELAY > FREEZE
SPACE	13	THROW, hold	CC 26 ch 13	B SPACE > SF REVERB > THROW
SPACE	14	FREEZE, toggle	note 25 (C#0) ch 13	B SPACE > SF REVERB > FREEZE
MACROS	15	ROLL, hold	CC 50 ch 14	20 DRUMS > SF REPEAT > ROLL

ADDRESSES

— **Channel = track:** 1 KICK, 2 BASS, 3 RUMBLE, 4 CLOSED, 5 OPEN, 6 CLAP, 7 STRIKE, 8 ACID, 9 STAB, 10 TEXTURE, 11 VOICE, 12 DUB, 13 SPACE, 14 the groups (LOW, DRUMS, MUSIC and DRUMS' REPEAT), 15 LIVE BUS and BRIDGE, 16 CONTROL.

— **CC = block base + the parameter's place in its plugin.** One plugin on a channel: base 20, so its performance row is CC 20-27. Channel 14: LOW ENERGY 20, DRUMS ENERGY 30, MUSIC ENERGY 40, REPEAT 50. Channel 15: BUILD 20. Channel 16: CONTROL 20.

— **Mixer:** CC 100 volume, 101 send A, 102 send B. Channel 14: 100-102 the LOW, DRUMS and MUSIC volumes; channel 15: 100 LIVE BUS, 101 BRIDGE.

— The plugins never reorder their parameters, so an address never moves. Every address is in the appendix.

LOAD THE SCENE

1. OXI App > Scenes. If no computer folder is assigned yet, assign one.
2. Copy **SF SHOW.oxie16** (controllers/e16 in the repository) into the Scenes folder inside it and click Refresh Views.
3. Drag SF SHOW from On Computer onto a free slot under On Device. The presses come with it.
4. **Auto Transmit off**, for the scene and in the global settings: sending stored values on a page change would pull Live's faders to whatever the E16 last held. MIDI Thru off.
5. In Live: E16 Port 1 Remote on, input and output; Ports 2 and 3 off (Port 1 is the one the E16 sends on; a second Remote port could take a message twice, and a FREEZE would flip twice). Takeover Mode: Value Scaling.

CHECK IT ONCE

1. Every knob: turn it, the right parameter moves. Every press: the right switch.
2. Hold DROP, a THROW and ROLL for three seconds each: on for the whole hold, off on release. (A long press also opens the E16's own functions; if one interferes, make that press a toggle.)
3. Shared knobs: on page 2 turn LOW ENERGY fully left; on page 1 turn it one step right. The sound moves one step, not jumps. If it jumps, the E16 keeps a value per page: controllers/e16 README,

check 4, has the relative fallback.

4. Change pages, leave SF SHOW and come back: Live's MIDI input indicator stays dark.
5. Press knob 1 (green) and turn: SLOW ALL moves, LOW ENERGY doesn't. Press again: LOW ENERGY.
6. Switch songs from the ONE: the rings follow, and a recalled knob moves smoothly from the new value.

5 THE PAGES

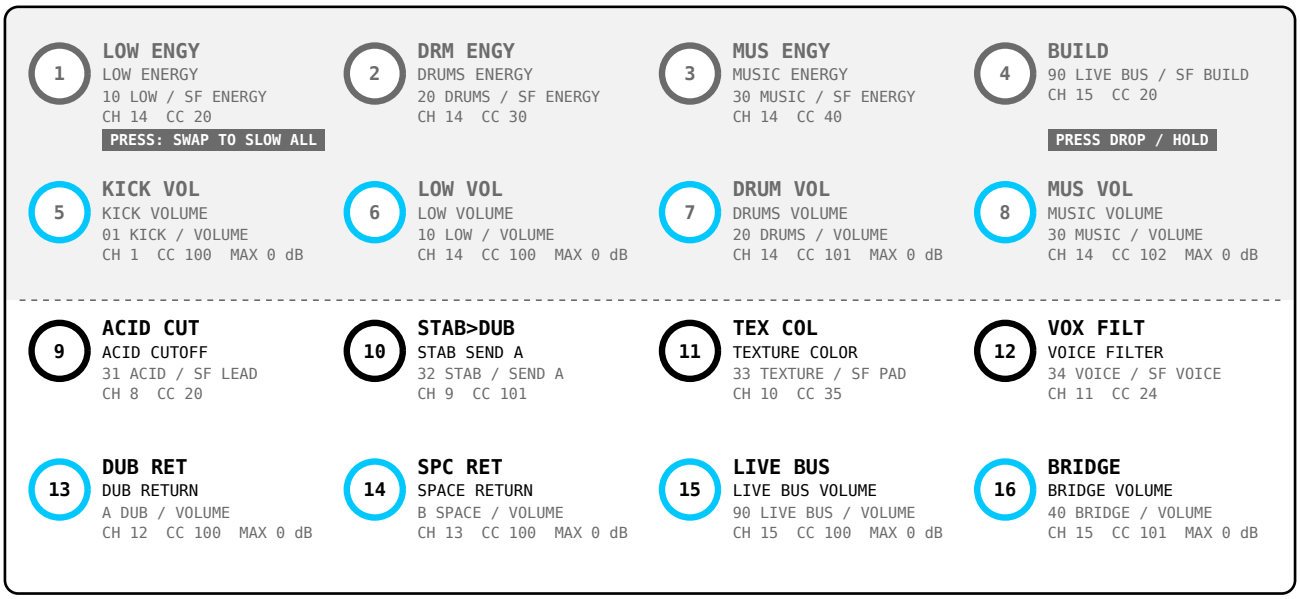
All twelve pages of SF SHOW. Rows 1 and 2 (grey) never change; each page is rows 3 and 4. Under each knob: the name on the E16's screen, the parameter, where it lands in Live, its address.

PAGE	ROW 3: KNOBS 9-12	ROW 4: KNOBS 13-16	PRESSES
1 PERFORM	ACID CUT / STAB>DUB / TEX COL / VOX FILT	DUB RET / SPC RET / LIVE BUS / BRIDGE	
2 KICK	TUNE / DECAY / PUNCH / CLICK	DENSITY / FILTER / DRIVE / LEVEL	
3 BASS	CUTOFF / RES / ENV AMT / GATE	DENSITY / FILTER / DRIVE / LEVEL	
4 RUMBLE	SIZE / DARK / GRIT / GATE	DENSITY / FILTER / PUMP / LEVEL	
5 HATS	CL DENS / CL DECAY / CL TONE / OP DENS	OP DECAY / OP TONE / CLP DENS / CLP DCY	
6 STRIKE	POWER / DISTANCE / SIZE / DECAY	DENSITY / FILTER / MATERIAL / LEVEL	
7 ACID	CUTOFF / RES / ENV AMT / GATE	DENSITY / FILTER / DRIVE / LEVEL	
8 STAB	CUTOFF / RES / ENV AMT / GATE	DENSITY / FILTER / SPACE / LEVEL	
9 TEXTURE	CUTOFF / COLOR / DETUNE / GATE	DENSITY / FILTER / ATTACK / LEVEL	
10 VOICE	SIGNAL / TONE / ECHO / GATE	DENSITY / FILTER / DRIVE / LEVEL	
11 SPACE	DUB TIME / DUB FDBK / DUB TONE / DUB DRV	SPC SIZE / SPC DCY / SPC DARK / SPC DUCK	9 THROW, 10 FREEZE, 13 THROW, 14 FREEZE
12 MACROS	MACRO 1 / MACRO 2 / MACRO 3 / MACRO 4	MACRO 5 / MACRO 6 / RPT LEN / RPT PTCH	15 ROLL

Knobs 1-8 are in chapter 4. On each page below: the drawing, what each of the page's eight knobs does (the plugin's own description), and how to play it.

PAGE 1 PERFORM

ROWS 3-4: THE MUSIC PARTS, THE RETURNS, LIVE BUS AND BRIDGE



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

The home page: the character of the four music parts, both returns, and the two big faders.

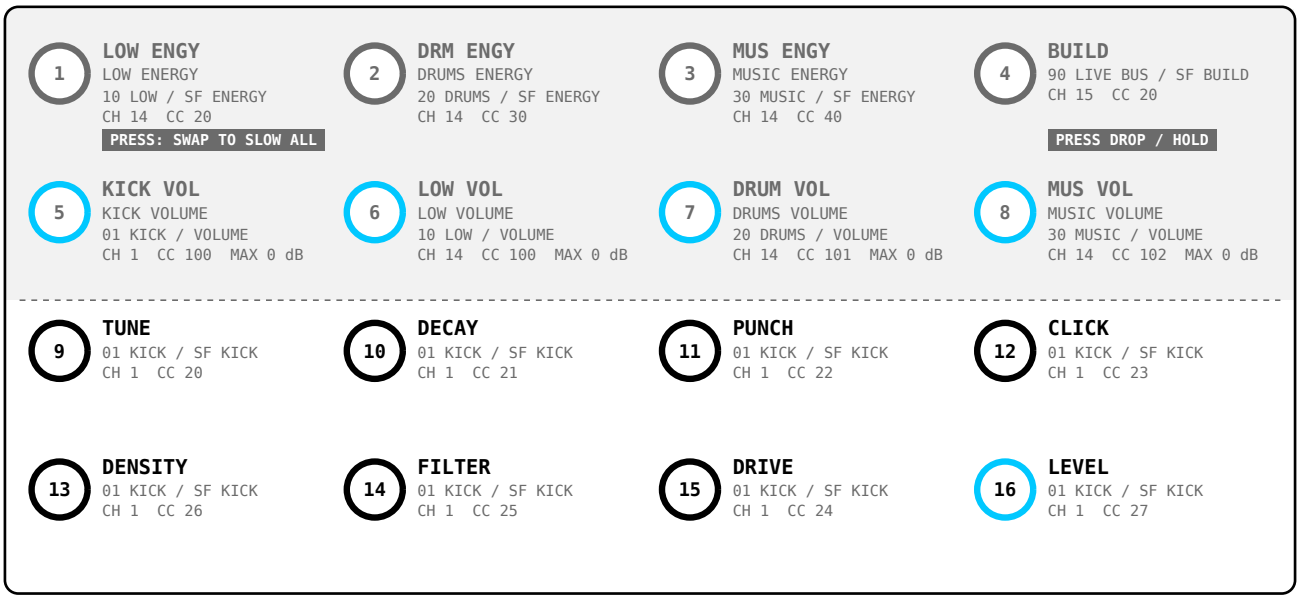
KNOB	SCREEN	WHAT IT DOES
9	ACID CUT	31 ACID: Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
10	STAB>DUB	Send from 32 STAB into A DUB, the dub delay. Sends are post-fader.
11	TEX COL	33 TEXTURE: Model-specific tone; on GRANULAR, the template's TEXTURE model, it is the grain size.
12	VOX FILT	34 VOICE: Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
13	DUB RET	The dub return's volume (SF DELAY on A DUB), up to 0 dB.
14	SPC RET	The reverb return's volume (SF REVERB on B SPACE), up to 0 dB.
15	LIVE BUS	The whole live set: every instrument and both returns. Pull it down to hand the room to BRIDGE.
16	BRIDGE	BRIDGE: your Live audio clips, for transitions and regrouping.

— 15 LIVE BUS and 16 BRIDGE are the escape hatch: bring BRIDGE up, pull LIVE BUS down, and the live set is gone while a BRIDGE clip plays. Regroup, then cross back.

— 10 sends STAB into DUB: the classic dub stab. Ride it with 13 DUB RETURN under the other hand.

PAGE 2 KICK

ROWS 3-4: SF KICK ON 01 KICK, CHANNEL 1



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF KICK on 01 KICK. TUNE and DECAY are the body, PUNCH and CLICK the attack.

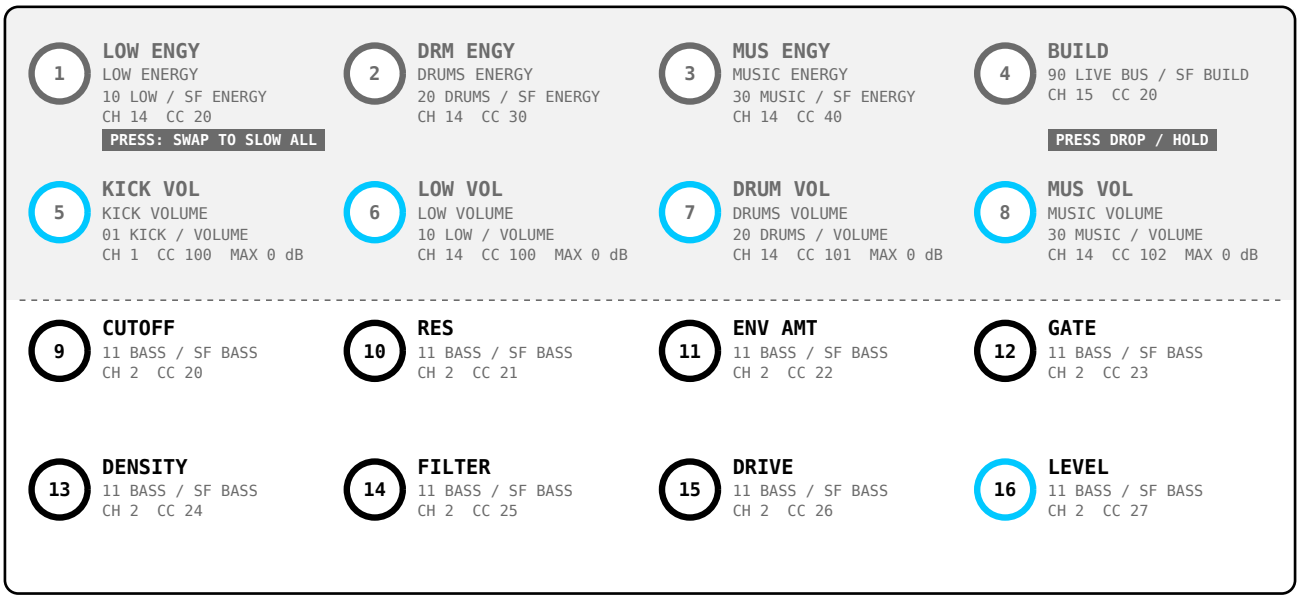
KNOB	SCREEN	WHAT IT DOES
9	TUNE	Kick pitch in semitones around the model's fundamental (about 44-55 Hz).
10	DECAY	Body length, x0.25 to x4 of the model's decay. This is the kick's GATE. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
11	PUNCH	Depth of the pitch sweep. More = harder transient, less = rounder.
12	CLICK	Level of the noise/impulse click on the attack.
13	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	DRIVE	Saturation on the kick, added to the model's own.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— Set TUNE and DECAY per song and store them: KICK and RUMBLE tune themselves to the song's KEY when KEY is on.

— FILTER right thins the kick, and Pocket closes the low pockets as it thins: the bass and rumble take the sub. Bring the kick's low end back on the drop.

PAGE 3 BASS

ROWS 3-4: SF BASS ON 11 BASS, CHANNEL 2



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF BASS on 11 BASS. CUTOFF, RES and ENV AMT move the line; GATE shortens it.

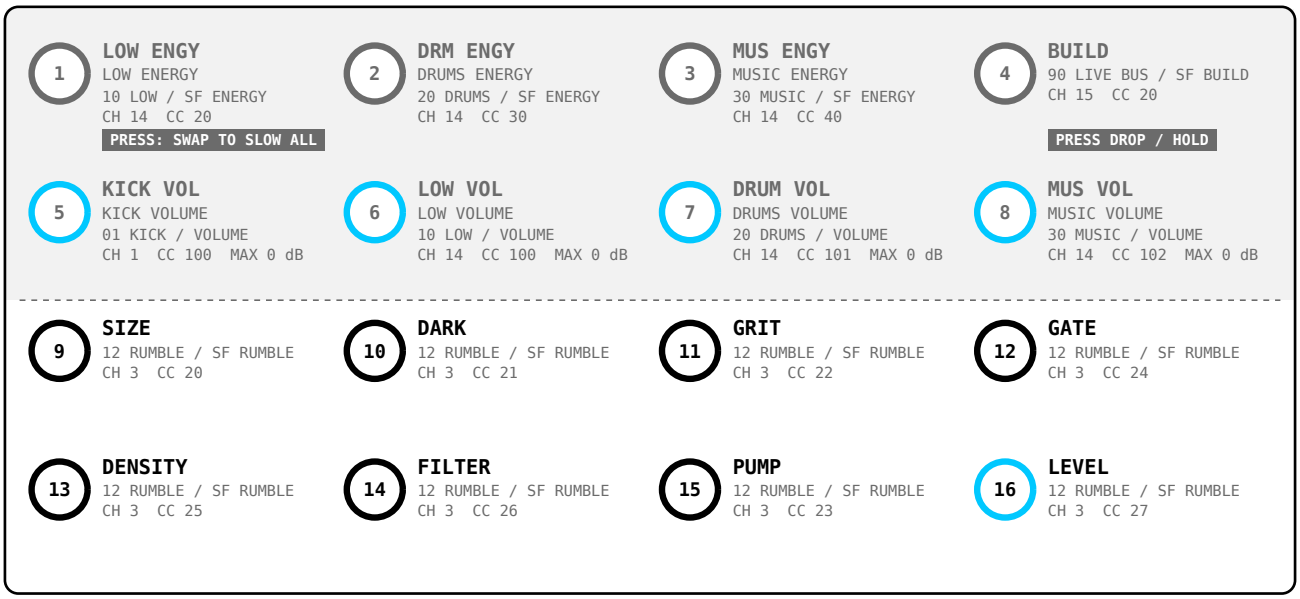
KNOB	SCREEN	WHAT IT DOES
9	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
10	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
11	ENV AMT	How far the filter envelope opens the filter on each note. Accented notes open further.
12	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
13	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	DRIVE	Saturation before the filter. Adds weight at low settings, grit at high.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— GATE down with DENSITY up is a tight, nervous build; GATE open with DENSITY down is spacious dub.

— DRIVE adds weight at low settings and grit at high ones.

PAGE 4 RUMBLE

ROWS 3-4: SF RUMBLE ON 12 RUMBLE, CHANNEL 3



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF RUMBLE on 12 RUMBLE. It plays no notes: it follows SF KICK.

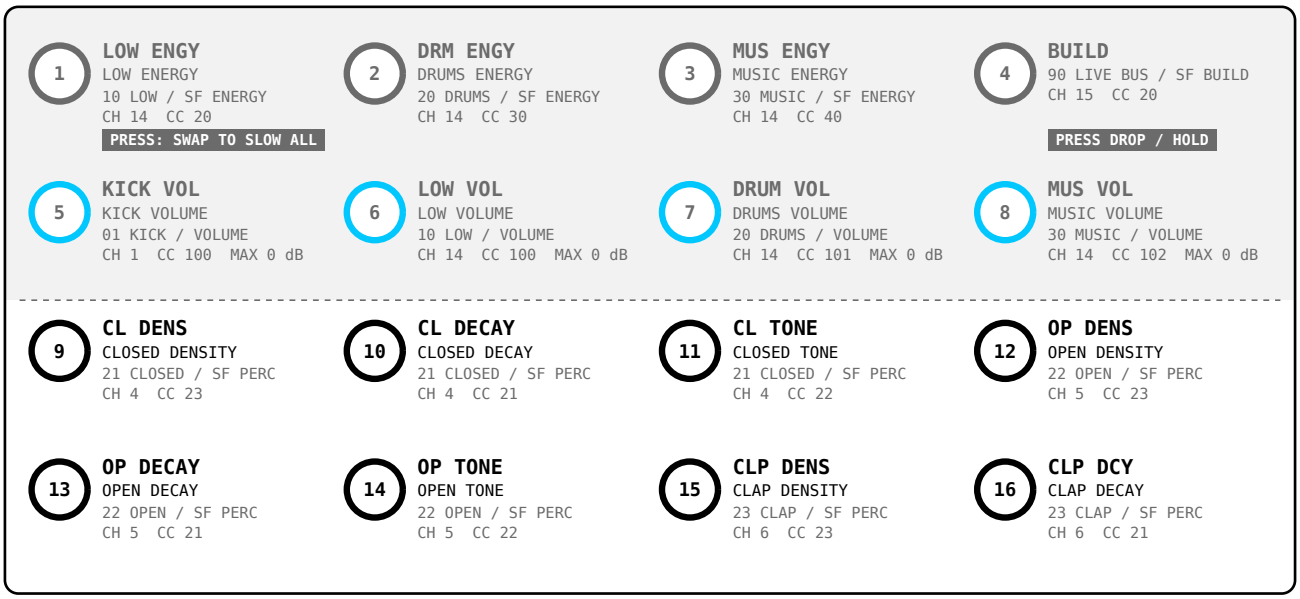
KNOB	SCREEN	WHAT IT DOES
9	SIZE	Reverb size / delay feedback.
10	DARK	Low-pass on the bed: 600 Hz down to 70 Hz.
11	GRIT	Distortion on the bed.
12	GATE	Cuts the tail after each kick. 100 % = no gate; lower = shorter.
13	DENSITY	Extra thumps between kicks (offbeat, then 16ths).
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	PUMP	Pocket depth: how far the bed ducks under each kick.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— SIZE is the bed's reverb, DARK its low-pass (600 Hz down to 70 Hz), GRIT its distortion; PUMP is how far it ducks under each kick.

— In each section decide whether BASS or RUMBLE carries the sustained low end, then steer that one.

PAGE 5 HATS

ROWS 3-4: SF PERC ON 21 CLOSED, 22 OPEN, 23 CLAP



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

Three SF PERCs on one page: CLOSED (9-11), OPEN (12-14), CLAP (15-16). The fill page.

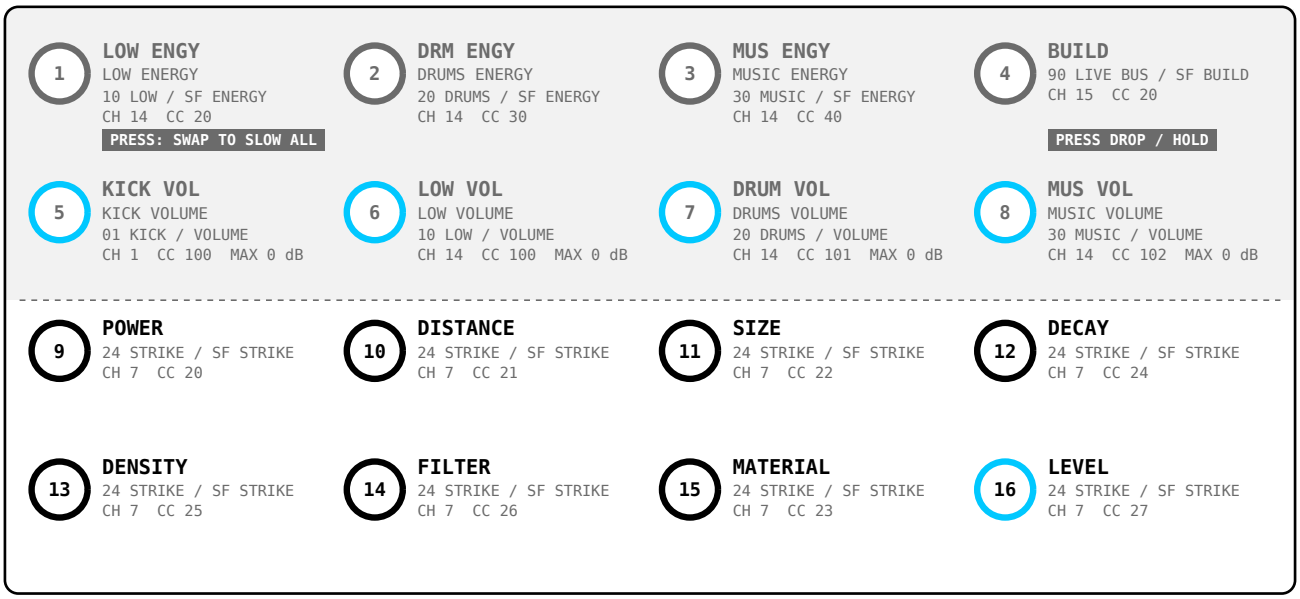
KNOB	SCREEN	WHAT IT DOES
9	CL DENS	21 CLOSED: Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
10	CL DECAY	21 CLOSED: Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
11	CL TONE	21 CLOSED: Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
12	OP DENS	22 OPEN: Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
13	OP DECAY	22 OPEN: Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
14	OP TONE	22 OPEN: Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
15	CLP DENS	23 CLAP: Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
16	CLP DCY	23 CLAP: Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.

— CLOSED DENSITY with ECHO SPACE at 1/16 turns 1/8 hats into 1/16 hats: the send-delay trick, built in.

— OPEN DECAY from closed to open across a build; CLAP DENSITY for rolls into the drop.

PAGE 6 STRIKE

ROWS 3-4: SF STRIKE ON 24 STRIKE, CHANNEL 7



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF STRIKE on 24 STRIKE: blast-excited bells, plates, bars, tubes and gongs.

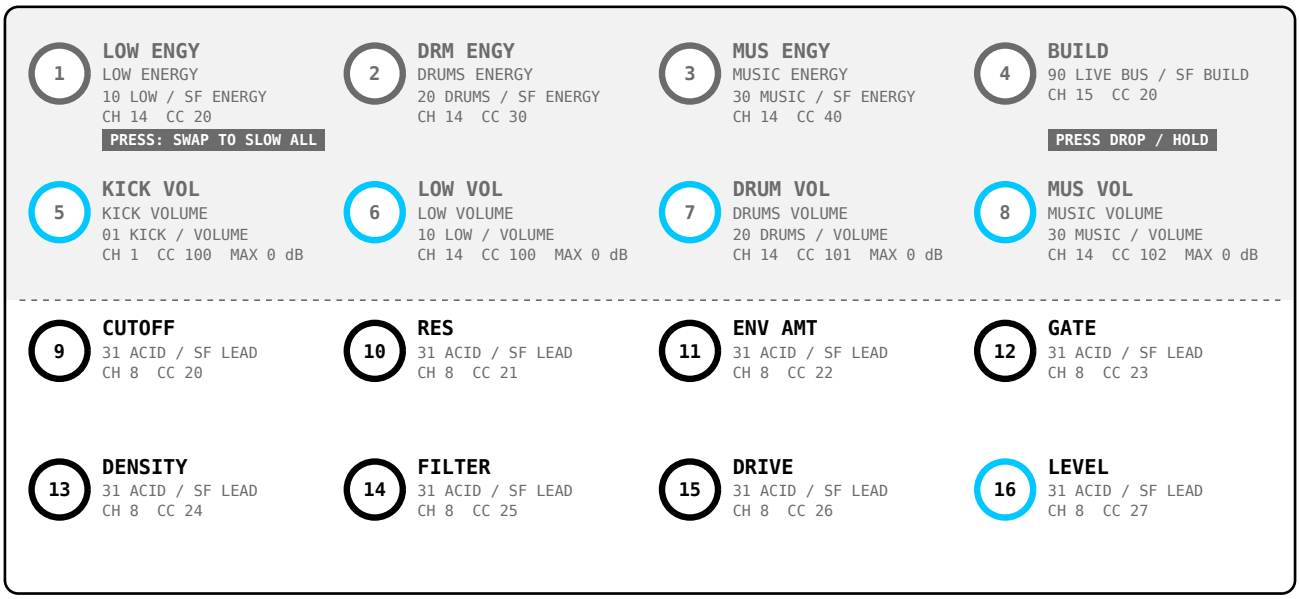
KNOB	SCREEN	WHAT IT DOES
9	POWER	Size of the charge. Velocity scales it (VEL SENS). Above about 70 % the object fractures.
10	DISTANCE	Stand-off. Further = quieter, longer, darker blast that excites fewer high modes.
11	SIZE	Object size, which sets the pitch: the fundamental goes from about 1.4 kHz (small) to 45 Hz (large).
12	DECAY	Ring time, 0.15 to 9 s, times the object's own (a gong rings longer than a bar). Four strikes ring at once; a fifth fades the quietest out over 5 ms.
13	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	MATERIAL	Wood (0) to glass (100 %): how long the high modes ring compared with the low ones.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— SIZE sets the pitch; MATERIAL goes from wood to glass; DISTANCE darkens and lengthens the hit.

— POWER above about 70 % fractures the object: save it for the peak.

PAGE 7 ACID

ROWS 3-4: SF LEAD ON 31 ACID, CHANNEL 8



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF LEAD (ACID 303-TYPE) on 31 ACID. CUTOFF, RES and ENV AMT are the squelch.

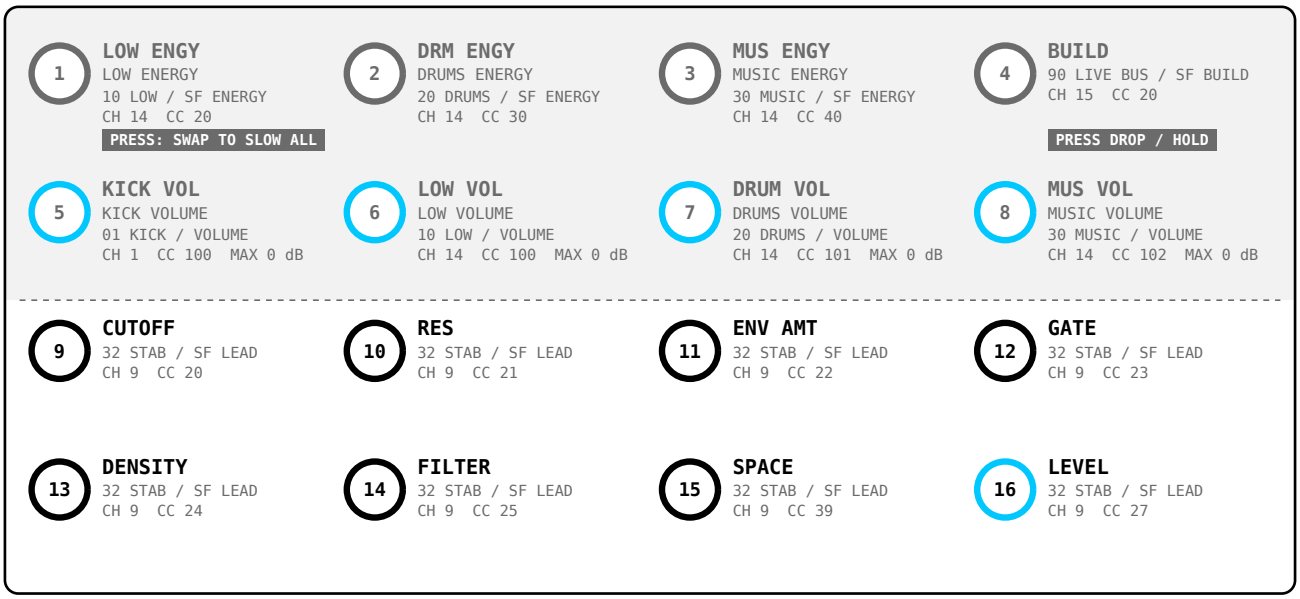
KNOB	SCREEN	WHAT IT DOES
9	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
10	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
11	ENV AMT	How far the filter envelope opens the filter on each note. Accented notes open further.
12	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
13	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	DRIVE	Saturation before the filter. Adds weight at low settings, grit at high.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— Overlapping notes slide like a 303 (always 60 ms): write legato notes on the ONE for slides.

— ACID CUTOFF also sits on PERFORM (9), so the acid line is never more than one hand away.

PAGE 8 STAB

ROWS 3-4: SF LEAD ON 32 STAB, CHANNEL 9



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF LEAD (STAB DUB) on 32 STAB. 15 SPACE is the stab's ensemble.

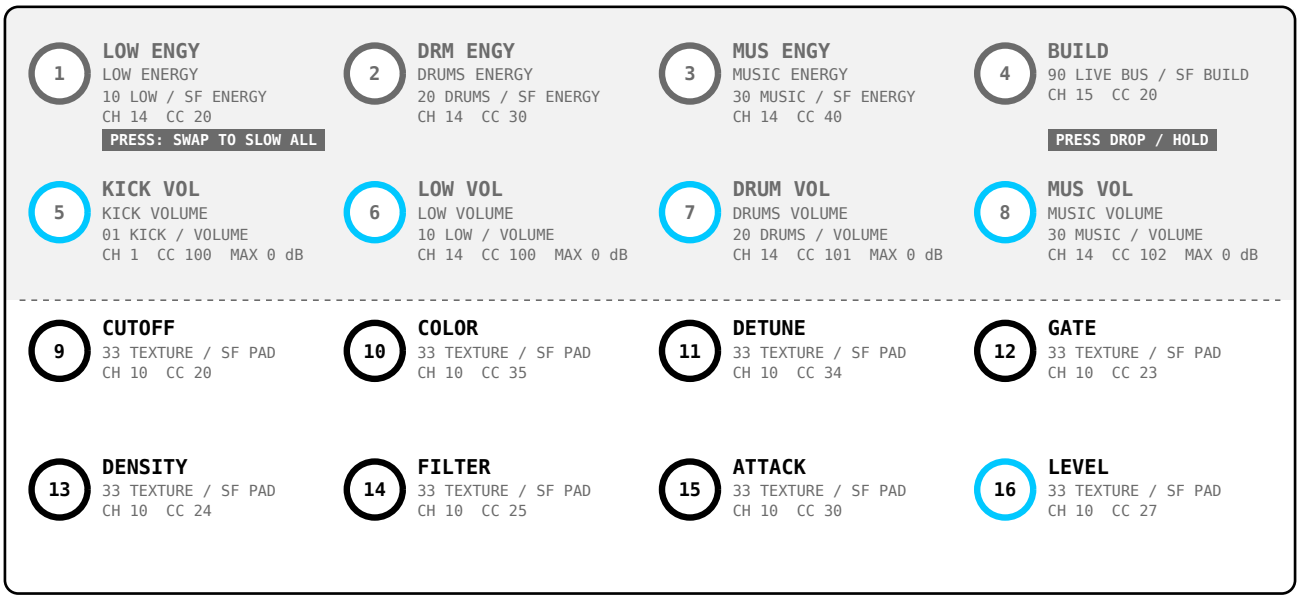
KNOB	SCREEN	WHAT IT DOES
9	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
10	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
11	ENV AMT	How far the filter envelope opens the filter on each note. Accented notes open further.
12	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
13	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	SPACE	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— If the ONE sends whole chords, set CHORD to NONE on the plugin; if it sends single notes, CHORD voices them. Not both.

— Short GATE plus DUB send (PERFORM 10) is the dub-chord move.

PAGE 9 TEXTURE

ROWS 3-4: SF PAD ON 33 TEXTURE, CHANNEL 10



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF PAD (GRANULAR) on 33 TEXTURE. On GRANULAR, COLOR is grain size and DETUNE is scatter.

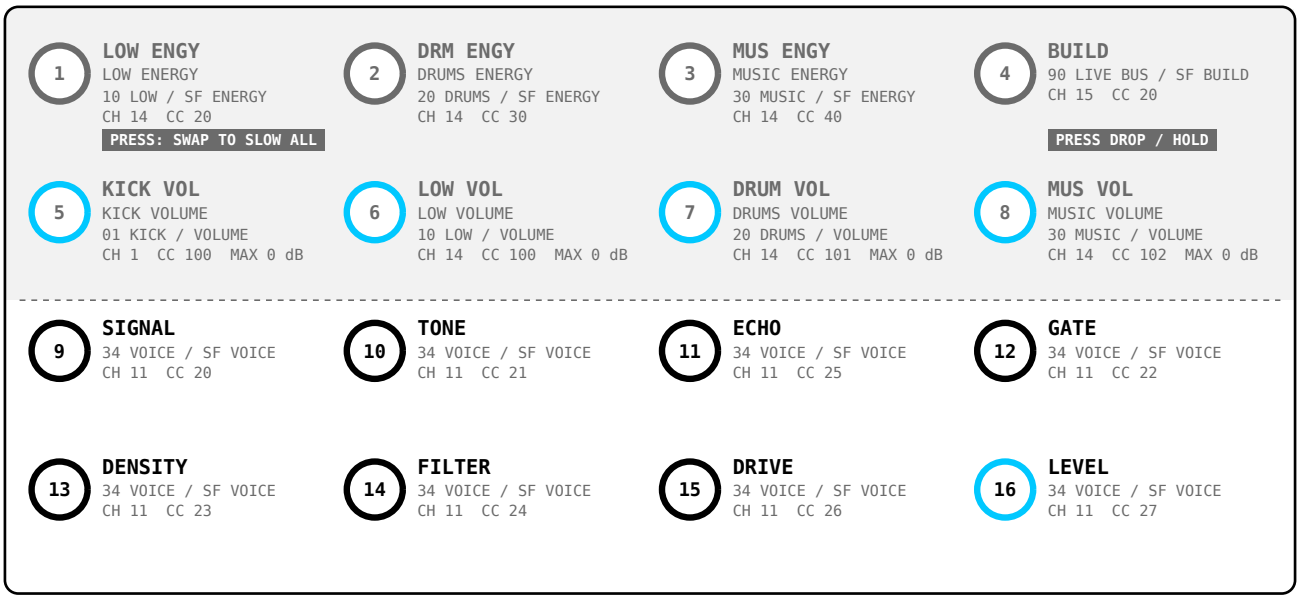
KNOB	SCREEN	WHAT IT DOES
9	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
10	COLOR	Model-specific tone; on GRANULAR, the template's TEXTURE model, it is the grain size.
11	DETUNE	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
12	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
13	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	ATTACK	Adds attack time to amp and filter envelopes. Up for swells.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— ATTACK up for swells; TEXTURE COLOR also sits on PERFORM (11).

— Field recordings work well in GRANULAR: drop one on the plugin before the show.

PAGE 10 VOICE

ROWS 3-4: SF VOICE ON 34 VOICE, CHANNEL 11



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF VOICE on 34 VOICE: phrases, radio fragments, choir.

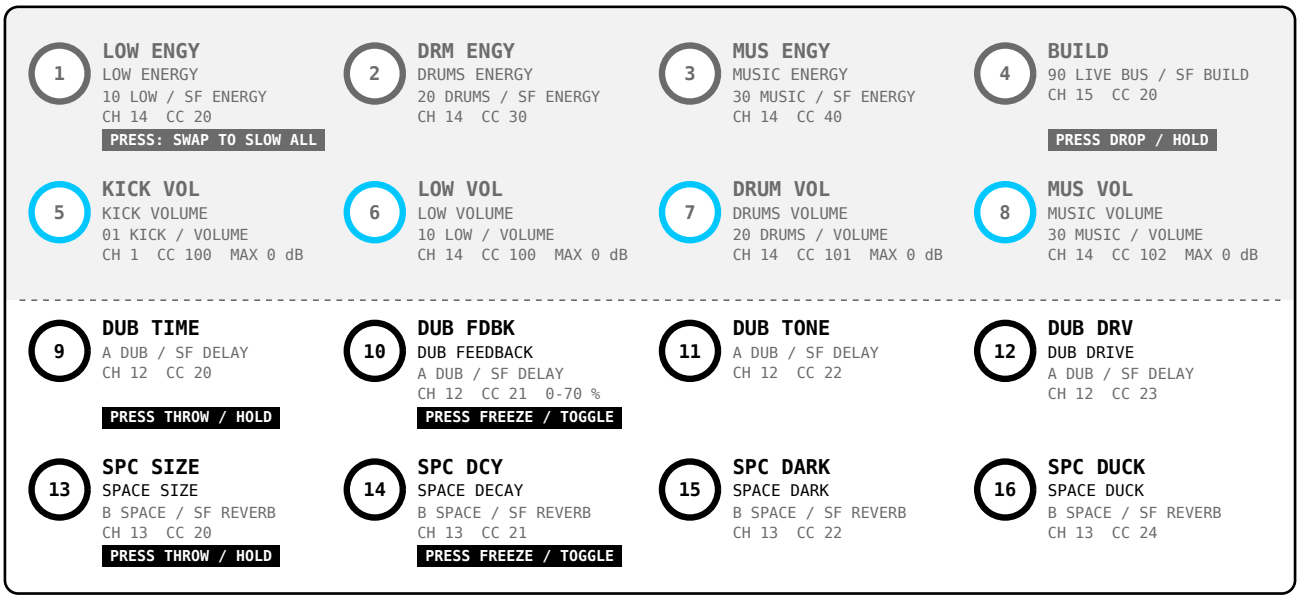
KNOB	SCREEN	WHAT IT DOES
9	SIGNAL	Clarity. RADIO: from lost to clear. Vocoders: sibilance and high-band brightness.
10	TONE	Model-specific: carrier pitch C1-C4 (vocoder, robot, choir), loop length (STUTTER), vowel (choir synth).
11	ECHO	Send into the built-in dub echo.
12	GATE	Slice length. Low values clip each slice short.
13	DENSITY	Re-triggers the slice as ghost repeats (stutter fills).
14	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
15	DRIVE	Saturation on the voice.
16	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.

— SIGNAL goes from lost to clear (RADIO model); ECHO feeds the voice's own dub echo.

— C3 plays slice 1, C#3 slice 2 and so on; below C3 the whole phrase. Sequence it as choreography.

PAGE 11 SPACE

ROWS 3-4: SF DELAY ON A DUB, SF REVERB ON B SPACE



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

The returns. Row 3 is DUB (SF DELAY on A DUB), row 4 is SPACE (SF REVERB on B SPACE). The presses sit under their knobs.

KNOB	SCREEN	WHAT IT DOES
9	DUB TIME	A DUB: Delay time, synced: 1 BAR to 1/32, then 5/16 and 7/16 (odd lengths: the repeats walk across the beat). Changing it slews the head (pitch bend).
10	DUB FDBK	A DUB: Repeats, up to 115 %. Above 100 % the loop saturates instead of growing.
11	DUB TONE	A DUB: Bipolar filter inside the feedback loop: left darker, right thinner. Each repeat goes further than the last.
12	DUB DRV	A DUB: Saturation inside the feedback loop (2x oversampled): each repeat dirtier than the last.
13	SPC SIZE	B SPACE: Room size.
14	SPC DCY	B SPACE: Decay time of the lows, 0.3 to 30 s (measured: DECAY is the real RT60).
15	SPC DARK	B SPACE: Damping: shortens the decay of the highs only, so the tail gets darker without getting shorter.
16	SPC DUCK	B SPACE: Ducks the tail under the kick and while the dry signal plays.

— THROW (hold) sends into the effect for as long as you hold it. FREEZE (toggle) loops what is in it until pressed again.

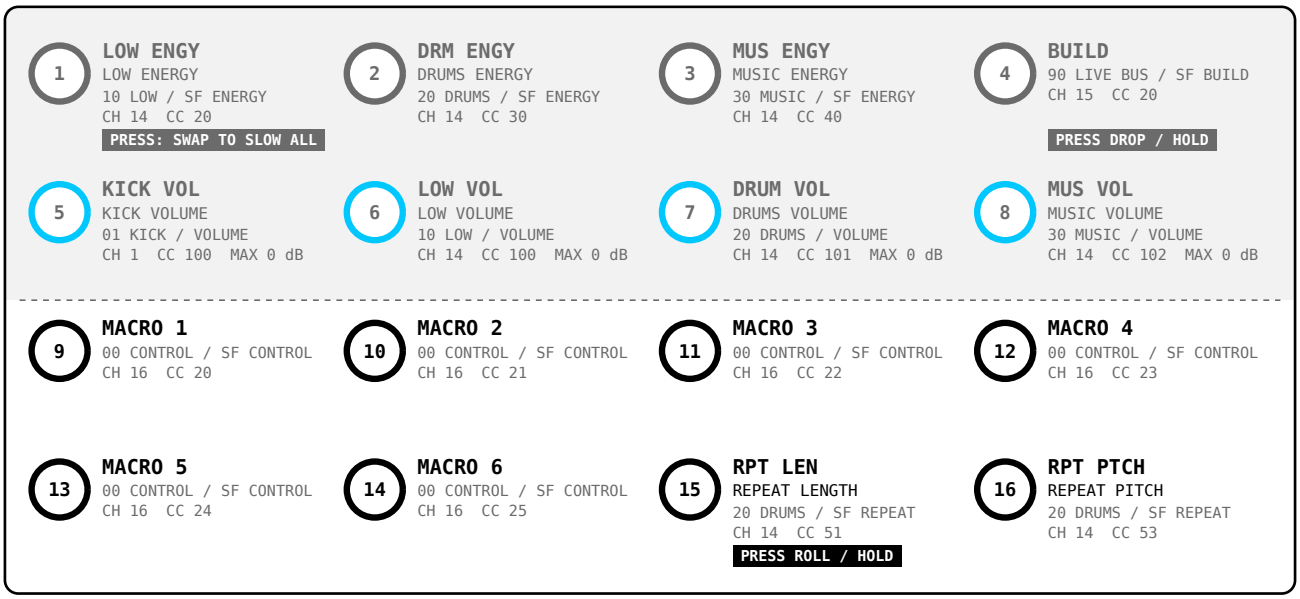
— Turn DUB TIME while frozen: the loop bends like tape speed and locks again when you let go.

— DUB FEEDBACK is mapped 0-70 % at first; widen the mapping when the music asks.

— SF TERROIR (your captured places) can replace SF REVERB on B SPACE: the same four knobs and two presses, mapped once (Cmd-M).

PAGE 12 MACROS

ROWS 3-4: SF CONTROL MACROS, SF REPEAT ON 20 DRUMS



GREY BAND: ROWS 1-2, THE SAME ON EVERY PAGE. RING: BLACK = A PARAMETER, LIGHT BLUE = A VOLUME. CH / CC: THE ADDRESS.

SF CONTROL's six macros, and SF REPEAT on 20 DRUMS.

KNOB	SCREEN	WHAT IT DOES
9	MACRO 1	Macro 1: every SF plugin with MACRO SRC = MACRO 1 follows it. ENERGY in the starting plan: the three ENERGY plugins and BUILD together.
10	MACRO 2	Macro 2: every SF plugin with MACRO SRC = MACRO 2 follows it. MOOD: TEXTURE, STAB and VOICE (cutoff, envelope, drive; the voice's SIGNAL and ECHO).
11	MACRO 3	Macro 3: every SF plugin with MACRO SRC = MACRO 3 follows it. DENSITY: CLOSED, OPEN, CLAP and STRIKE (DECAY and DENSITY: the fills).
12	MACRO 4	Macro 4: every SF plugin with MACRO SRC = MACRO 4 follows it. PULSE: KICK and RUMBLE (kick DECAY, PUNCH, DRIVE; rumble SIZE, GRIT, DENSITY).
13	MACRO 5	Macro 5: every SF plugin with MACRO SRC = MACRO 5 follows it. SPACE: the DUB and SPACE returns (delay FEEDBACK and MIX; reverb SIZE, DECAY and MIX).
14	MACRO 6	Macro 6: every SF plugin with MACRO SRC = MACRO 6 follows it. FRACTURE: ACID and BASS (cutoff, envelope, density and drive: smooth to heavy).
15	RPT LEN	20 DRUMS: Loop length: 1/4 to 1/32 and triplets.
16	RPT PTCH	20 DRUMS: Pitch drop over the repeats: falls by this amount every four passes.

— Hold ROLL on 15 and turn LENGTH under the same finger: the drums roll at the length you dial.

— 16 PITCH makes the roll fall as it repeats. The macro plan is in chapter 7.

6 SONGS, PATTERNS AND SCENES

Each system owns one thing. The ONE owns the notes, SF songs own the sound, the E16 owns where your hands are. A song change is one Program Change: every SF plugin lands on the same bar.

SYSTEM	RESPONSIBLE FOR
OXI ONE projects and patterns	Notes, rhythms, gates, accents, the pattern progression, and the Program Change that names the song.
SF songs (SF CONTROL)	The sound of every SF plugin and the song's KEY, recalled on the bar. Sixteen songs, saved with the set.
E16 pages	Where your hands are. Pages hold no sound: changing pages changes nothing you hear.
Live Session scenes	BRIDGE clips, reference material, rehearsal.
Live Arrangement	Recording the performance.

A SONG CHANGE



— SF CONTROL decides the bar line once and sends it with the song, so every plugin lands on the same bar whatever order Live processes the tracks in.

— A change sent within 30 ms after a bar line (a Program Change on the downbeat) counts for that bar; later in the bar, for the next one. Every plugin's header shows WAIT BAR in green until it lands.

— Turning SONG on SF CONTROL does the same as a Program Change. A plugin with nothing stored in that song stays as it is.

— **SLOW ALL** on SF CONTROL makes a recall glide: at 8 or 16 BAR every SF knob the song stores travels to it over 8 or 16 bars; at OFF it lands on the bar. Menus and switches (a MODEL, an ECHO SPACE) land on the bar either way (GLIDES, chapter 7, has the list).

SAVING A SONG

— Set every plugin how the song should start, then press **STORE ALL** on SF CONTROL: every SF plugin stores its values into the song in force (SONG).

— **Into another song:** choose it in **STORE TO** first, then STORE ALL. That song isn't recalled, so nothing you hear changes; it becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG. Choosing it on SONG instead would recall it and lose what you set.

— If a song change is under way when you STORE TO (WAIT BAR), the change still lands and stays the song in force.

— STORE in a plugin's header stores just that plugin. Nothing that rewrites a song is on the E16.

A TRACK IN THE SET

1. Load the track's ONE project: sequencer 8 sends its Program Change, and the song lands on the bar.
2. Play its patterns in order (SPARSE, MAIN, VARIATION, EXIT) from the ONE; play the sound from the E16.

3. Before the next track: BUILD and a drop, a dub throw or a BRIDGE clip covers the change (chapter 7).

7 PERFORMANCE

The top half is the arrangement, the page is the detail. Very simple note content, all the variation played on knobs: every move below is a knob or a press, on the bar.

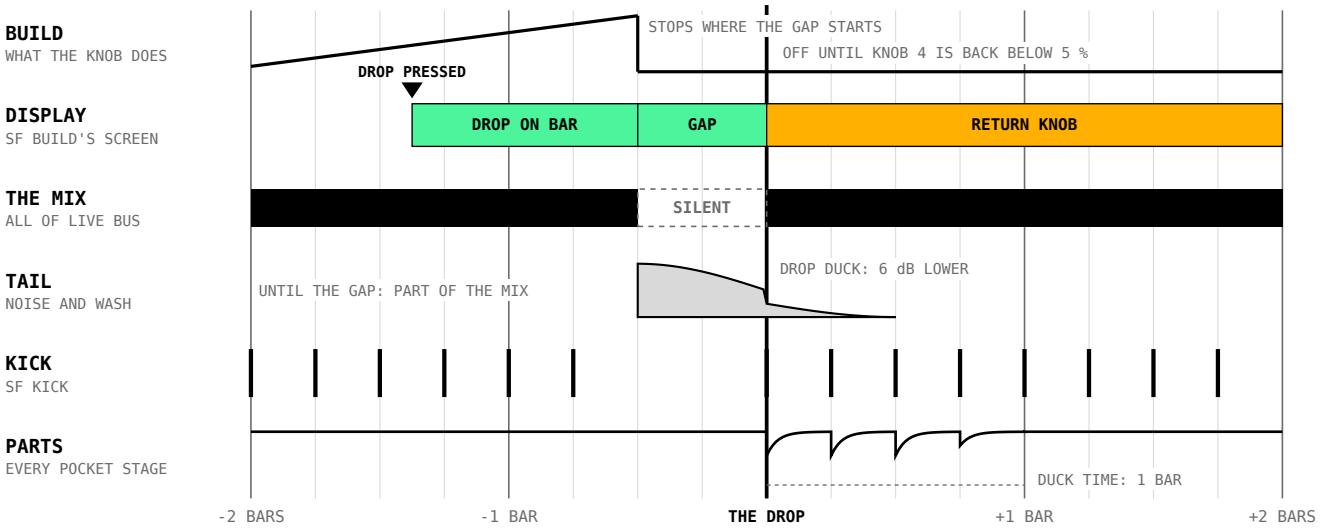
THE SET ARC

PHASE	WHAT CARRIES IT	WHERE
OPENING	Kick and rumble alone; TEXTURE or VOICE far back. ENERGY knobs left of centre.	Rows 1-2; RUMBLE, TEXTURE
LOCK-IN	Hats come in by DENSITY, not by level. One part at a time, eight or sixteen bars apart.	HATS, KICK
BUILD	BUILD up over 8-16 bars; hats and clap DENSITY up; SPACE DUCK down so the tail fills the bar.	Knob 4, HATS, SPACE
PEAK	The drop; ENERGY knobs right of centre; acid or stab carries the hook; dub throws on the offbeats.	Press 4, PERFORM, ACID, SPACE
BREAKDOWN	Kick lane muted on the ONE; LOW down; FREEZE a return; VOICE up.	The ONE, rows 1-2, SPACE, VOICE
RETURN	Kick back on the one; a song change lands here.	Press 4, the ONE
CLOSE	ENERGY knobs left; parts out one by one; the last bars on a frozen reverb or BRIDGE.	Rows 1-2, SPACE, PERFORM 16

A starting plan, not a rule: hypnotic sets hold a phase far longer than feels safe.

BUILDS AND DROPS

BUILD is one knob for a whole-mix build: a rising high-pass, a noise riser, a reverb wash and width. Press DROP (knob 4) during the build: the drop lands on the next bar line. Three settings on SF BUILD shape the landing; the drawing shows GAP 1/2, TAIL 1 BAR, DROP DUCK 6 dB, DUCK TIME 1 BAR.



— **GAP** (OFF to 1 BAR): everything through BUILD goes silent for that long before the bar line, the whole LIVE BUS with its returns (BRIDGE and DECK play on). Press DROP later than that and the gap starts on the next 1/16.

— **TAIL** (CUT to 2 BARS): the riser's noise and the wash ring on after the build stops, fading out. Shorter than the GAP, it dies away in the silence; longer, it carries into the drop, DROP DUCK lower.

— **DROP DUCK** (0-12 dB, off by default) and **DUCK TIME** (1/4 to 4 BARS): after the drop every SF part with Pocket dips under each kick, whatever its POCKET DEPTH: bass, lead, pad, voice, rumble, both returns. The first kicks stand alone on top of everything that comes back, with less weight than a kick brought back by hand: the bass and rumble around them are gone. A roll into the drop doesn't duck; the duck starts with the first kick on the bar.

— The build lets go 10 ms before the bar line, the gap's silence with it, so a kick that reaches Live a little early keeps its attack.

— After a drop the build stays off until knob 4 is back below 5 % (display: RETURN KNOB). Bring it down in your own time: nothing re-builds.

RECIPE	GAP	TAIL	DROP DUCK	DUCK TIME	WHAT IT DOES
HARD STOP	1/4	CUT	OFF	-	A beat of true silence, then everything on the one.
BREATH	1/2	1/4	OFF	-	The riser rings into the silence and dies before the kick.
CARRY	OFF	1 BAR	6 dB	1 BAR	No silence: the wash runs under the first bar, ducked.
LONG FALL	1 BAR	1/2	9 dB	2 BARS	A full bar of air; the new section breathes for two bars.
SNAP	OFF	CUT	OFF	-	The build snaps off on the bar line; nothing else changes.

Defaults: GAP OFF, TAIL 1/8, DROP DUCK OFF, DUCK TIME 1 BAR. The drop duck needs KICK REPORTING on SF CONTROL. TAP and STRIP set to ROLE KICK, PERC or DECK don't duck.

ENERGY AND MACROS

— The three ENERGY knobs are the fastest way to move a whole set: small moves, often. A negative MACRO 1 amount on LOW darkens the bass while the rest lifts.

— Six macros on the MACROS page, each followed by the plugins that subscribe to it (MACRO SRC, MACRO AMT on each plugin). A starting plan:

MACRO	FOLLOWED BY	MOVES
1 ENERGY	the three ENERGY plugins, BUILD	every group's energy at once
2 MOOD	TEXTURE, STAB, VOICE	cutoff, envelope and drive; the voice's SIGNAL and ECHO
3 DENSITY	CLOSED, OPEN, CLAP, STRIKE	DECAY and DENSITY: the fills
4 PULSE	KICK, RUMBLE	kick DECAY, PUNCH, DRIVE; rumble SIZE, GRIT, DENSITY
5 SPACE	DUB, SPACE returns	delay FEEDBACK and MIX; reverb SIZE, DECAY and MIX
6 FRACTURE	ACID, BASS	cutoff, envelope, density and drive: smooth to heavy

Start the amounts at 30-50 %. One macro per plugin.

DUB

— **Throws:** SPACE page, hold 9 (DUB THROW) on an offbeat stab or a voice fragment and let go: the repeats carry on. With INPUT THROW ONLY on SF DELAY, the press is the only way in.

— **Tape moves:** turn DUB TIME while repeats are running: the head slews and the pitch bends. FREEZE (10) then TIME: the frozen loop bends like tape speed.

— **Repeats that grow:** DUB FEEDBACK past 100 % saturates instead of exploding; the mapping stops at 70 % until you widen it.

— **Space that breathes:** SPACE DUCK 30-60 % lets the reverb bloom in the gaps between kicks. FREEZE (14) holds a chord in the room for a breakdown.

FILLS AND TRANSITIONS

— **Hats:** HATS page, CLOSED DENSITY up with ECHO SPACE at 1/16 on SF PERC: 1/8 hats become 1/16s, then 1/32 rolls past 50 %.

— **Drum roll:** MACROS page, hold ROLL (15) and turn LENGTH under the same finger: 1/4 down to 1/32. PITCH (16) makes it fall.

— **Between tracks:** load the next ONE project during a build or a breakdown; the song lands on a bar line. With SLOW ALL set, the knobs glide into the new song; a MODEL change in it still lands on the bar, so cover it with the build or the drop.

— **BRIDGE:** bring 16 BRIDGE up on PERFORM and 15 LIVE BUS down to hand the room to a clip while you regroup. BRIDGE still needs Live: it is not a backup for a computer failure.

GLIDES

— **A travelling knob's ring turns green** and walks with the sound to where you set it, then turns white. Turn it while it walks: the ring shows your hand, then green again. (SF CONTROL's E16 RINGS and the scene's script; nothing else depends on them.)

— **SLOW ALL** (press knob 1, turn, press again) sets how long your next moves take, in every SF plugin. OFF: they land at once. 8 BAR: a quarter turn of ACID CUTOFF becomes an eight-bar rise that plays itself.

— **A move keeps its length.** Start a 16-bar rise on the pad, set SLOW ALL to OFF, and make quick moves elsewhere: the pad keeps rising. Turn a travelling knob and it sets off again from where it has got to.

— **SLOW ALL changed in the same instant as a move** (one audio buffer, a few ms): the move takes the length from before the change, in every plugin alike. Set SLOW ALL, then move.

— **Song changes follow it, knob by knob.** SLOW ALL at 16 BAR before the Program Change: every SF knob the song stores travels to the new song over 16 bars; OFF: it lands on the bar. What does not glide (the table) changes on the bar. It is a morph of the knobs, not a crossfade between two whole sounds.

— A plugin's own SLOW, set in its window, wins over SLOW ALL; leave it at CONTROL.

WHAT SLOW MOVES (every knob and press of SF SHOW; drawn from the scene and the plugins)

ON THE E16	WITH SLOW / SLOW ALL
Every SF knob, 90 of them: each plugin's top row, and the details on the scene too (STAB SPACE, TEXTURE ATTACK, TEXTURE COLOR, TEXTURE DETUNE)	Travels over the bars set
SLOW ALL, DUB TIME, REPEAT LENGTH	Menus: at once, on the next buffer
DROP, THROW, FREEZE, ROLL	Presses: at once, as long as held or until pressed again
KICK VOLUME, LOW VOLUME, DRUMS VOLUME, MUSIC VOLUME, STAB SEND A, DUB RETURN, SPACE RETURN, LIVE BUS VOLUME, BRIDGE VOLUME	Live's own faders and send: never glide
Not on the E16: MODEL (an instrument's sound)	At once. What sounds releases on the old model; the keys still down play again on the new one (a held chord goes on)
Not on the E16: CHORD, a dropped sample, any other menu or switch	At once: CHORD from the next note; a sample replaces the old one, notes sounding included

WHEN SOMETHING GOES WRONG

WHAT HAPPENED	DO THIS
A runaway: feedback, a stuck note, a wrong sound in front of the crowd	The interface's output down first: the room goes quiet, REC keeps recording. Then PANIC, held in SF CONTROL's window: every SF plugin fades out over 5 ms and clears; let go, it comes back without a click.
A plugin header shows FAULT in red	It caught an invalid value or a runaway and reset itself; it carries on. Note what was playing.
The wrong song landed	Send the right Program Change (or turn SONG): it lands on the next bar.
A knob seems dead	Another page is up. Or its range in Live's MIDI Mappings is narrow.
The ONE drifts or stops	Live is the clock: stop and start Live, the ONE follows. Check its USB cable before the show, not during it.
The ONE doesn't follow Live's tempo	Start from Live, not the ONE: Live sends clock only while it plays. The ONE's header shows USB (stopped: Shift + Stop until it does). Sync on OXI ONE MKII (Port 1)'s output and on no other port. Hold Shift + Back for 2 s: the ONE's MIDI monitor shows the clock arriving while Live plays. Still nothing: Sync on Port 2's output instead, never both.

8 BEFORE THE SHOW

Run through this list at every soundcheck. Each line is a thing that once went wrong somewhere.

CHECKLIST

1. Every SF plugin from the same build: SF CONTROL shows BUS SHARED and KICK REPORTING, and the pump lands on the kick.
2. The ONE's Program Change recalls the song: every track on the same bar, and a recalled knob is yours the moment you turn it.
3. The interface's output down: the room silent, REC still recording.
4. The PA pair carries 92 PA and nothing else; Cue reaches the headphones and not the PA.
5. DROP, a THROW and ROLL each held for three seconds: on for the whole hold, off on release.
6. The shared E16 knobs don't jump (chapter 4); changing pages sends nothing.
7. SLOW ALL where you want it for the opening (knob 1 pressed; OFF for direct moves), then back to LOW ENERGY.
8. A drop rehearsed with the night's GAP, TAIL and DROP DUCK settings.
9. A minute recorded on REC with a deep ROOM EQ cut: the cut isn't in the recording.
10. Soundcheck the busiest passage: SF LIMIT INPUT for 1-3 dB of limiting there, AUTO on, DOWN ONLY.
11. The room mics on 94 ROOM: IN L and IN R move on its SF CAPTURE, the same gain on both, well under the top at the loudest.

CAPTURING THE ROOM

SF TERROIR can take the place you play in as a PLACE, without leaving Live: SF CAPTURE on 94 ROOM plays the sweep to a speaker on the headphone out and records the room mics, and TERROIR turns the take into a place. SF TERROIR has to be in the set (on B SPACE in place of SF REVERB: controllers/e16 README, SF TERROIR ON B SPACE). At soundcheck or after the set: the transport stopped.

1. A speaker on the headphone out (a MiniRig on its line in), where the sound should come from, aimed at a wall or into the room. In TotalMix the mics must not reach the headphone out, or the speaker plays them and feeds back.
2. The mics where the audience will be, or where the place should be heard from (SF TERROIR's IN THE FIELD, MICS). The speaker as loud as it goes without buzzing; the headphone level sets it, SWEEP LEVEL on SF CAPTURE is only for headroom.
3. Stop the transport. In SF TERROIR click **CAPTURE** (amber), then the slot to fill. 46 s: keep the room quiet while the status line counts.
4. TERROIR analyses the take: the slot takes the name ROOM and the date and time; the status line should read SWEEP x3 and 30 dB or more over the noise. Under 30 dB: the speaker louder or the mics closer; THE MICS CLIPPED: the speaker or the gain down. Then again. THE SWEEP DIDN'T COME BACK: nothing reached the mics (TROUBLESHOOTING); the slot stays as it was.
5. PLACE is stored per song, so a recall goes back to the slot the song was stored with: STORE the songs that should play the new place. The take stays in Music/SF Live Suite/TERROIR/Captures.

CAPTURE again (it reads STOP) ends a take early; so do starting the transport, PANIC and Live's audio restarting, even before the sweep has started or while the take is being written. The sweep fades out in 10 ms and nothing is stored. The PA stays silent: 94 ROOM goes to the headphone out only.

RECORDING THE SET

— Press RESET on SF LIMIT, arm the instrument tracks, 93 REC and 94 ROOM, Automation Arm on, Arrangement record, play. The stereo recording on REC is the set, the one on ROOM the room mics for the mix; the MIDI and automation are for editing and rehearsal.

— Collect All and Save after the show. SF samples are inside the set; DECK tracks are referenced by path, so keep the files where they are.

TROUBLESHOOTING

SYMPTOM	CHECK
An instrument doesn't play	MIDI From: its ONE port and channel, never All Ins. The ONE's sequencer output (1A, 2B ...) matches.
Every instrument plays every part	A track's MIDI From is All Ins or All Channels.
Nothing ducks	SF CONTROL: BUS SHARED and KICK REPORTING. The kick must be SF KICK.
Pump early or late	Live buffer 512 or less; the ONE's MIDI Clock Sync Delay; then POCKET OFFSET by ear.
Songs don't change	PC ENABLE on, PC CHANNEL 16, sequencer 8 on 16B. The VST3 needs Live 12.2 or later; otherwise use the AU.
E16 rings don't follow	Remote on for E16 Port 1's output.
An E16 press does nothing	An SF SHOW file from before 3 Nov is loaded (it had no presses): load the current one. A toggle that flips back: Remote is on for two E16 ports.
No green on a travelling knob	SF CONTROL shows E16 RINGS PORT 3 in green? If not, the E16 isn't found: try PORT 1. Load SF SHOW from the repository again (the script is in it). Use the VST3s.
Knob 1 swapped (green) changes nothing	SLOW ALL isn't mapped in this set: Cmd-M, click SLOW ALL on SF CONTROL, press knob 1 and turn it. Cmd-M off, save.
Knob 1 moves LOW ENERGY and SLOW ALL together	Its push was changed in the App: load SF SHOW.oxie16 again (knob 1's push must be Swap Destination).
Every move glides (or none does)	SLOW ALL is saved with the set: press knob 1 and set it. A plugin whose own SLOW isn't CONTROL keeps its own.
An E16 knob jumps when touched	Auto Transmit is on, or the E16 keeps a value per page: chapter 4, check 3.
CONTROL shows LOCAL ONLY	The plugins come from different builds: rebuild and install the whole suite, restart Live.
A return doubles the dry signal	DRY off on SF DELAY or SF REVERB on the returns.
SF TERROIR won't capture	Its status line says why: the transport playing; NO SF CAPTURE (94 ROOM's device off); a take already running.
A capture says THE SWEEP DIDN'T COME BACK, or under 30 dB	The sweep didn't reach the speaker or the mics: the speaker on the headphone out, 94 ROOM on and up; Input Config 1/2 a stereo pair, Monitor In. Then louder or closer. The slot stays as it was.
A capture says NO SWEEP IN THE RIGHT CHANNEL	That mic heard nothing (its cable, its gain): the place is mono. Fix it, capture again.
A capture says SF CAPTURE STOPPED ANSWERING	94 ROOM's SF CAPTURE was removed or switched off. Put it back, capture again: nothing was stored.

APPENDIX

E16 ADDRESS MAP (SF SHOW)

Every address of the scene once, by channel: the Live target, the mapping, and where the knob is on the E16. The same list is controllers/e16/sf-show-map.csv.

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
1	CC 20	01 KICK > SF KICK > TUNE	absolute	KICK 9
1	CC 21	01 KICK > SF KICK > DECAY	absolute	KICK 10
1	CC 22	01 KICK > SF KICK > PUNCH	absolute	KICK 11
1	CC 23	01 KICK > SF KICK > CLICK	absolute	KICK 12
1	CC 24	01 KICK > SF KICK > DRIVE	absolute	KICK 15
1	CC 25	01 KICK > SF KICK > FILTER	absolute	KICK 14
1	CC 26	01 KICK > SF KICK > DENSITY	absolute	KICK 13
1	CC 27	01 KICK > SF KICK > LEVEL	absolute	KICK 16
1	CC 100	01 KICK > mixer > volume	absolute, Max 0 dB	every page, encoder 5
2	CC 20	11 BASS > SF BASS > CUTOFF	absolute	BASS 9
2	CC 21	11 BASS > SF BASS > RES	absolute	BASS 10
2	CC 22	11 BASS > SF BASS > ENV AMT	absolute	BASS 11
2	CC 23	11 BASS > SF BASS > GATE	absolute	BASS 12
2	CC 24	11 BASS > SF BASS > DENSITY	absolute	BASS 13
2	CC 25	11 BASS > SF BASS > FILTER	absolute	BASS 14
2	CC 26	11 BASS > SF BASS > DRIVE	absolute	BASS 15
2	CC 27	11 BASS > SF BASS > LEVEL	absolute	BASS 16
3	CC 20	12 RUMBLE > SF RUMBLE > SIZE	absolute	RUMBLE 9
3	CC 21	12 RUMBLE > SF RUMBLE > DARK	absolute	RUMBLE 10
3	CC 22	12 RUMBLE > SF RUMBLE > GRIT	absolute	RUMBLE 11
3	CC 23	12 RUMBLE > SF RUMBLE > PUMP	absolute	RUMBLE 15
3	CC 24	12 RUMBLE > SF RUMBLE > GATE	absolute	RUMBLE 12
3	CC 25	12 RUMBLE > SF RUMBLE > DENSITY	absolute	RUMBLE 13
3	CC 26	12 RUMBLE > SF RUMBLE > FILTER	absolute	RUMBLE 14
3	CC 27	12 RUMBLE > SF RUMBLE > LEVEL	absolute	RUMBLE 16
4	CC 21	21 CLOSED > SF PERC > DECAY	absolute	HATS 10
4	CC 22	21 CLOSED > SF PERC > TONE	absolute	HATS 11
4	CC 23	21 CLOSED > SF PERC > DENSITY	absolute	HATS 9
5	CC 21	22 OPEN > SF PERC > DECAY	absolute	HATS 13
5	CC 22	22 OPEN > SF PERC > TONE	absolute	HATS 14
5	CC 23	22 OPEN > SF PERC > DENSITY	absolute	HATS 12
6	CC 21	23 CLAP > SF PERC > DECAY	absolute	HATS 16
6	CC 23	23 CLAP > SF PERC > DENSITY	absolute	HATS 15
7	CC 20	24 STRIKE > SF STRIKE > POWER	absolute	STRIKE 9
7	CC 21	24 STRIKE > SF STRIKE > DISTANCE	absolute	STRIKE 10
7	CC 22	24 STRIKE > SF STRIKE > SIZE	absolute	STRIKE 11

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
7	CC 23	24 STRIKE > SF STRIKE > MATERIAL	absolute	STRIKE 15
7	CC 24	24 STRIKE > SF STRIKE > DECAY	absolute	STRIKE 12
7	CC 25	24 STRIKE > SF STRIKE > DENSITY	absolute	STRIKE 13
7	CC 26	24 STRIKE > SF STRIKE > FILTER	absolute	STRIKE 14
7	CC 27	24 STRIKE > SF STRIKE > LEVEL	absolute	STRIKE 16
8	CC 20	31 ACID > SF LEAD > CUTOFF	absolute	PERFORM 9, ACID 9
8	CC 21	31 ACID > SF LEAD > RES	absolute	ACID 10
8	CC 22	31 ACID > SF LEAD > ENV AMT	absolute	ACID 11
8	CC 23	31 ACID > SF LEAD > GATE	absolute	ACID 12
8	CC 24	31 ACID > SF LEAD > DENSITY	absolute	ACID 13
8	CC 25	31 ACID > SF LEAD > FILTER	absolute	ACID 14
8	CC 26	31 ACID > SF LEAD > DRIVE	absolute	ACID 15
8	CC 27	31 ACID > SF LEAD > LEVEL	absolute	ACID 16
9	CC 20	32 STAB > SF LEAD > CUTOFF	absolute	STAB 9
9	CC 21	32 STAB > SF LEAD > RES	absolute	STAB 10
9	CC 22	32 STAB > SF LEAD > ENV AMT	absolute	STAB 11
9	CC 23	32 STAB > SF LEAD > GATE	absolute	STAB 12
9	CC 24	32 STAB > SF LEAD > DENSITY	absolute	STAB 13
9	CC 25	32 STAB > SF LEAD > FILTER	absolute	STAB 14
9	CC 27	32 STAB > SF LEAD > LEVEL	absolute	STAB 16
9	CC 39	32 STAB > SF LEAD > SPACE	absolute	STAB 15
9	CC 101	32 STAB > mixer > send A	absolute	PERFORM 10
10	CC 20	33 TEXTURE > SF PAD > CUTOFF	absolute	TEXTURE 9
10	CC 23	33 TEXTURE > SF PAD > GATE	absolute	TEXTURE 12
10	CC 24	33 TEXTURE > SF PAD > DENSITY	absolute	TEXTURE 13
10	CC 25	33 TEXTURE > SF PAD > FILTER	absolute	TEXTURE 14
10	CC 27	33 TEXTURE > SF PAD > LEVEL	absolute	TEXTURE 16
10	CC 30	33 TEXTURE > SF PAD > ATTACK	absolute	TEXTURE 15
10	CC 34	33 TEXTURE > SF PAD > DETUNE	absolute	TEXTURE 11
10	CC 35	33 TEXTURE > SF PAD > COLOR	absolute	PERFORM 11, TEXTURE 10
11	CC 20	34 VOICE > SF VOICE > SIGNAL	absolute	VOICE 9
11	CC 21	34 VOICE > SF VOICE > TONE	absolute	VOICE 10
11	CC 22	34 VOICE > SF VOICE > GATE	absolute	VOICE 12
11	CC 23	34 VOICE > SF VOICE > DENSITY	absolute	VOICE 13
11	CC 24	34 VOICE > SF VOICE > FILTER	absolute	PERFORM 12, VOICE 14
11	CC 25	34 VOICE > SF VOICE > ECHO	absolute	VOICE 11
11	CC 26	34 VOICE > SF VOICE > DRIVE	absolute	VOICE 15
11	CC 27	34 VOICE > SF VOICE > LEVEL	absolute	VOICE 16
12	CC 20	A DUB > SF DELAY > TIME	absolute	SPACE 9
12	CC 21	A DUB > SF DELAY > FEEDBACK	absolute, 0-70 % at first	SPACE 10
12	CC 22	A DUB > SF DELAY > TONE	absolute	SPACE 11
12	CC 23	A DUB > SF DELAY > DRIVE	absolute	SPACE 12

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
12	CC 24	A DUB > SF DELAY > THROW	press: hold	SPACE 9
12	CC 100	A DUB > mixer > volume	absolute, Max 0 dB	PERFORM 13
12	NOTE 25 (C#0)	A DUB > SF DELAY > FREEZE	press: toggle	SPACE 10
13	CC 20	B SPACE > SF REVERB > SIZE	absolute	SPACE 13
13	CC 21	B SPACE > SF REVERB > DECAY	absolute	SPACE 14
13	CC 22	B SPACE > SF REVERB > DARK	absolute	SPACE 15
13	CC 24	B SPACE > SF REVERB > DUCK	absolute	SPACE 16
13	CC 26	B SPACE > SF REVERB > THROW	press: hold	SPACE 13
13	CC 100	B SPACE > mixer > volume	absolute, Max 0 dB	PERFORM 14
13	NOTE 25 (C#0)	B SPACE > SF REVERB > FREEZE	press: toggle	SPACE 14
14	CC 20	10 LOW > SF ENERGY > ENERGY	absolute	every page, encoder 1
14	CC 30	20 DRUMS > SF ENERGY > ENERGY	absolute	every page, encoder 2
14	CC 40	30 MUSIC > SF ENERGY > ENERGY	absolute	every page, encoder 3
14	CC 50	20 DRUMS > SF REPEAT > ROLL	press: hold	MACROS 15
14	CC 51	20 DRUMS > SF REPEAT > LENGTH	absolute	MACROS 15
14	CC 53	20 DRUMS > SF REPEAT > PITCH	absolute	MACROS 16
14	CC 100	10 LOW > mixer > volume	absolute, Max 0 dB	every page, encoder 6
14	CC 101	20 DRUMS > mixer > volume	absolute, Max 0 dB	every page, encoder 7
14	CC 102	30 MUSIC > mixer > volume	absolute, Max 0 dB	every page, encoder 8
15	CC 20	90 LIVE BUS > SF BUILD > BUILD	absolute	every page, encoder 4
15	CC 21	90 LIVE BUS > SF BUILD > DROP	press: hold	every page, encoder 4
15	CC 100	90 LIVE BUS > mixer > volume	absolute, Max 0 dB	PERFORM 15
15	CC 101	40 BRIDGE > mixer > volume	absolute, Max 0 dB	PERFORM 16
16	CC 20	00 CONTROL > SF CONTROL > MACRO 1	absolute	MACROS 9
16	CC 21	00 CONTROL > SF CONTROL > MACRO 2	absolute	MACROS 10
16	CC 22	00 CONTROL > SF CONTROL > MACRO 3	absolute	MACROS 11
16	CC 23	00 CONTROL > SF CONTROL > MACRO 4	absolute	MACROS 12
16	CC 24	00 CONTROL > SF CONTROL > MACRO 5	absolute	MACROS 13
16	CC 25	00 CONTROL > SF CONTROL > MACRO 6	absolute	MACROS 14
16	CC 34	00 CONTROL > SF CONTROL > SLOW ALL	absolute	every page, encoder 1, second destination